Runbook: AWS EC2 Elastic IP Migration and Association

1. Purpose

This document explains how to:

1 Move an existing Elastic IP from an old EC2 server (Server A) to a new EC2 server (Server B).

2 Create a new Elastic IP and associate it with an EC2 instance.

An Elastic IP is a fixed public IPv4 address. Unlike an automatically assigned public IP, it remains the same while it is allocated to your AWS account.

2. Current Setup

- **Server A:** Existing EC2 server with an Elastic IP.
- **Server B:** New EC2 server with an automatically assigned public IP.

Example

Detail	Server A	Server B
Before migration	Existing server	New server
Public IP	Elastic IP: `13.200.10.20`	Auto-assigned IP: `3.110.20.30`
After migration	No longer associated with the Elastic IP	Receives `13.200.10.20`

Important: The IP addresses above are examples only.

3. Prerequisites

Before starting, confirm the following:

[ ] You have access to the AWS Management Console.

[ ] Server B is in the `Running` state.

[ ] The application is installed and working on Server B.

[ ] Required files, data, and database connectivity are available.

[ ] Security groups allow the required traffic.

[ ] The correct AWS Region is selected.

[ ] The Elastic IP address and instance IDs are recorded.

[ ] Production changes have the required approval.

[ ] Server A will remain available until migration validation is complete.

4. Option A: Move the Existing Elastic IP from Server A to Server B

Step 1: Verify Server B

1 Sign in to the AWS Management Console.

2 Open **EC2 › Instances**.

3 Select Server B.

4 Confirm that the instance state is `Running`.

5 Test the application using Server B's current public IP, where supported.

6 Verify that the application, database connectivity, and required services work correctly.

Expected result: Server B is ready to receive production traffic.

> Do not continue if the application is not working correctly.

Step 2: Identify the Existing Elastic IP

1 Open **EC2 › Network & Security › Elastic IP addresses**.

2 Find the Elastic IP associated with Server A.

3 Record the Elastic IP address and allocation ID.

4 Verify that the associated instance ID belongs to Server A.

Expected result: The correct Elastic IP has been identified.

Step 3: Associate the Existing Elastic IP with Server B

1 Select the existing Elastic IP.

2 Click **Actions › Associate Elastic IP address**. If the console presents a reassociation option, use that option instead.

3 Select `Instance` as the resource type, if prompted.

4 Select Server B's instance ID.

5 Select the correct private IP address if prompted.

6 Review the details.

7 Click **Associate** to confirm.

Expected result: The existing Elastic IP is now associated with Server B instead of Server A.

> **Important:** Do not select **Release Elastic IP address**.

Step 4: Verify the Migration

1 Open **EC2 › Instances**.

2 Select Server B.

3 Check its public IPv4 address.

4 Confirm that it matches the original Elastic IP.

5 Return to **Elastic IP addresses**.

6 Verify that the Elastic IP is associated with Server B's instance ID.

7 Test the application using the existing public IP address.

8 Test the domain name, if applicable.

Expected result: Users can access the application through the same public IP address, now directed to Server B.

If the domain already points to the Elastic IP, a DNS change is generally unnecessary.

5. Option B: Create a New Elastic IP and Attach It to an EC2 Instance

Use this option when you need a new fixed public IP instead of transferring the existing Elastic IP.

Step 1: Open Elastic IP Addresses

1 Sign in to the AWS Management Console.

2 Open **EC2**.

3 Select **Network & Security › Elastic IP addresses**.

4 Click **Allocate Elastic IP address**.

Step 2: Allocate a New Elastic IP

1 Keep the default Amazon IPv4 address pool selected unless your requirements specify otherwise.

2 Review the configuration.

3 Click **Allocate**.

4 Record the new Elastic IP address and allocation ID.

Expected result: AWS allocates a new Elastic IP to your account.

Step 3: Associate the New Elastic IP with an EC2 Instance

1 Select the newly allocated Elastic IP.

2 Click **Actions › Associate Elastic IP address**.

3 Select `Instance` as the resource type.

4 Choose the EC2 instance to which you want to attach the IP.

5 Select the appropriate private IP address if prompted.

6 Review the details.

7 Click **Associate**.

Expected result: The new Elastic IP is associated with the selected EC2 instance.

Step 4: Verify the New Elastic IP

1 Open **EC2 › Instances**.

2 Select the target instance.

3 Check the public IPv4 address.

4 Confirm that it matches the new Elastic IP.

5 Test application access using the new IP address.

6 Update DNS records if the domain must point to this new IP.

Expected result: The EC2 instance can be accessed through the new Elastic IP, provided the application and network configuration allow it.

6. Rollback Procedure for Option A

If Server B fails after the migration and you need to restore Server A:

1 Confirm that Server A is available and ready to serve traffic.

2 Open **EC2 › Network & Security › Elastic IP addresses**.

3 Select the same Elastic IP.

4 Choose **Actions › Associate Elastic IP address** or the available reassociation option.

5 Select Server A's instance ID.

6 Select the appropriate private IP address if prompted.

7 Click **Associate**.

8 Verify the association with Server A.

9 Test the application through the existing public IP address or domain name.

10 Notify the relevant team of the rollback result.

Expected result: The existing Elastic IP points back to Server A.

> Rollback can cause downtime. Keep Server A available until Server B has been validated successfully.

7. Troubleshooting

Problem	What to check
Application does not open	Check the application service, security groups, network ACLs, and required ports.
Elastic IP is not shown on the target instance	Verify the Elastic IP association and instance ID.
SSH connection fails	Check SSH port access, security group rules, routing, and credentials.
Domain does not open	Check DNS records, SSL certificates, and web server configuration.
Application opens but data is missing	Check database connectivity, storage, configuration, and data migration.
AWS does not allow association	Verify the AWS Region, selected private IP, and IAM permissions.

8. Completion Checklist

[ ] The correct Elastic IP has been associated with the target EC2 instance.

[ ] The instance is running.

[ ] The application is accessible through the intended public IP.

[ ] Domain access is working, if applicable.

[ ] Application and database functions have been tested.

[ ] Logs and monitoring have been checked.

[ ] The team has been notified of the result.

[ ] The old server has been retained until migration approval and validation.

9. Important Notes

1 Moving an Elastic IP does not copy application files, databases, or other data between servers.

2 The target instance must be properly configured before the IP is moved.

3 Moving an Elastic IP can interrupt existing connections and cause temporary downtime.

4 Creating a new Elastic IP does not transfer the old IP.

5 AWS may charge for public IPv4 addresses. Check current pricing before allocating additional addresses.

6 Do not release an Elastic IP that is still required.

7 Do not terminate the old server until the migration is confirmed successful and decommissioning is approved.

10. Final Outcome

Option A (Existing Elastic IP): The same public IP is moved from Server A to Server B. This is useful when users must continue accessing the application through the existing IP address.

Option B (New Elastic IP): A new fixed public IP is created and attached to an EC2 instance. If a domain is used, its DNS record may need to be updated.

Recommendation: For a production migration, use an approved maintenance window, validate the new server before switching traffic, and keep the old server available until the migration is confirmed successful.